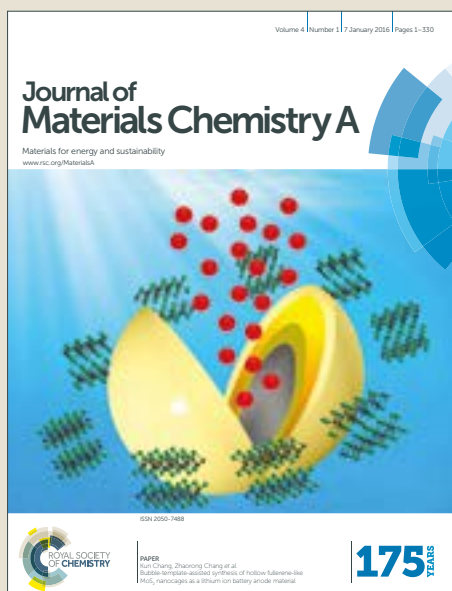


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Journal of Materials Chemistry A

A green and scalable route to yield porous carbon sheets from biomass for supercapacitors with high capacity

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Abstract:

A green and scalable route is developed to prepare hierarchical porous carbon sheets (HPCS) from biomass (cornstalk) directly under air atmosphere instead of inert gas protection. In addition, the method adopts non-toxic NaCl and KCl mixed salt as reaction media to regulate the activation process, which avoids the corrosive and poisonous KOH and ZnCl₂ reagents. Meanwhile, the pore structures of HPCS can be well-manipulated by the salt templates and the etching effects of the oxygen atom as well as the high energy molten salt media. The as-obtained HPCS consist of ultra-thin sheet structure (~4.6 nm in thickness) and abundant hierarchical pores which facilitates the electrolyte diffusion and ion transfer. In addition, the high specific surface area of 1588 m² g⁻¹ provides sufficient active sites and the rich O doping

introduces pseudo capacitance. Therefore, the as-prepared HPCS exhibit a high specific capacitance of 407 F g^{-1} at 1 A g^{-1} in a three-electrode system. Moreover, the assembled symmetrical supercapacitor in a two-electrode system also exhibits an outstanding specific capacitance of 413 F g^{-1} at 0.5 A g^{-1} , excellent rate capacity and a long-term cycling stability with 92.6 % of initial capacitance retention after 20 000 cycles at 5 A g^{-1} , which are comparable to that of the porous carbons activated with traditional methods. It is expected that this green and facile technique could serve as an effective and scalable method to prepare optimized HPCS with tunable pore structures from biomass for electrochemical energy storage application.

Keywords: Porous carbon; Biomass; Air atmosphere; Molten Salt; Supercapacitors

1. Introduction

Electrochemical energy storage has been playing a significant role in the development of clean and sustainable energy technologies.¹⁻⁶ As a class of electrochemical devices, supercapacitors are currently attracting intensive attention due to their excellent power density and long cycle life.⁷⁻¹⁵ Electrode materials are usually considered to be crucial for supercapacitors and among various electrode materials, carbon materials are the most widely used because of their outstanding advantages such as tunable pore architectures, large specific surface areas, good electronic conductivity and favorable chemical/thermal stability.¹⁶⁻²⁰

Electrical double layer capacitors (EDLC) with pure carbonaceous electrode materials have been commercial available but the low specific capacitance and energy density still hinder their applications. Capacitance of supercapacitors mainly arises from the interface reaction of electrode materials, therefore, increasing specific surface area of carbon materials can improve the electrochemical double-layered capacitive behavior of the electrodes by providing more

surface storage sites²¹. In addition, incorporation of heteroatom species such as nitrogen, oxygen etc. into carbon materials could also enhance the capacitance of carbon materials by inducing pseudocapacitance^{9, 10, 21}. However, for EDLC, the main challenge is to enhance the specific capacitance and the energy density while preserving high power and long cycling life. Therefore, optimizing electrode structure such as constructing two-dimensional thin carbon sheets for reducing diffusive resistance to mass transport on the large electrode/electrolyte interfaces²²⁻²⁶ is one promising approach. Meanwhile, developing hierarchical porous structures is another promising approach to form macro-, meso- and micro-pores respectively for ion-buffering reservoirs (macro-pores), ion-transporting channels (meso-pores) and ion-storage sites (micro-pores)²⁷⁻³². Template assisted methods were also employed to construct hierarchical porous carbon structures with high specific surface area³³⁻³⁷. Although they can effectively realize the control of pore structure, they usually suffered from complicate synthesis processes, expensive templates or carbon precursors, and poisonous template removing reagents, which greatly limits their practical applications.

Compared to unsustainable carbon resources, biomass can serve as a renewable resource to prepare porous carbons for the applications of electrochemical energy storage.³⁸⁻⁴⁶ To obtain porous carbon materials with large surface area from biomass, chemical activation methods (activated with KOH, ZnCl₂, etc) and molten salts are usually adopted⁴⁷⁻⁵⁰. However, these techniques usually suffer from toxic or strongly corrosive chemical reagent (ZnCl₂, KOH) to obtain the final products. Moreover, most activated carbon electrodes activated with KOH usually display narrow micro-pore characteristics, which limits the transport of large sized ions and undermines the performances of EDLC especially at high current densities²⁵. Additionally, up to now the synthesis of biomass-derived porous carbons is performed predominantly under

inert gas atmosphere (like N₂ and Ar), which requires large amounts of resource consumption as well as harsh technical and equipment requirements, hindering the commercial production. With respect to these concerns, therefore, it is very essential and highly needed to develop a green and facile large-scale synthesis technique to construct carbon electrodes from biomass satisfying all the aforementioned characteristics of carbonaceous materials for high-performance EDLC.

Herein, a green and scalable strategy is presented to fabricate hierarchical porous carbon sheets (HPCS) from cornstalk at 800 °C directly under air atmosphere. Non-toxic NaCl and KCl mixed salt is employed as the protection and activation media to prevent the carbon product from burning and regulate the pore structures of HPCS. Meanwhile, trace of oxygen in air also takes part in generating pores in HPCS. As a result, the as-obtained HPCS exhibit thin carbon sheet architecture with abundant multilevel hierarchical pores, high specific surface area of 1588 m² g⁻¹ and rich O doping. The thin carbon sheets facilitate rapid charge transfer⁴⁶, the large specific area surface offer sufficient active sites for charge capacity^{43,46}, adequate multilevel hierarchical pores facilitate the accessibility of the electrolyte access to the electrode surface^{46,55} and the rich O doping introduce pseudo-capacitance^{21, 43}. Therefore, adopted as electrode materials for supercapacitors, HPCS show a high specific gravimetric capacitance of 407 F g⁻¹ at 1 A g⁻¹ in a three-electrode system. In addition, the symmetrical supercapacitor assembled with HPCS also delivers a high specific gravimetric capacitance of 413 F g⁻¹ at 0.5 A g⁻¹, excellent rate capacity and long-term cycling stability with 92.6 % of initial capacitance retention after 20,000 cycles at 5 A g⁻¹ in a two-electrode system.

2. Experimental Section

2.1. Material Synthesis

Firstly, KCl was mixed with NaCl in a mass ratio of 1:1 by grinding in an agate mortar. The

cornstalk was peeled off and cornstalk cores were cut into pieces. Then 3, 2, and 1g pieced cornstalk cores were immersed in 50ml saturated salt solution for 12 h, respectively. After immersion, the cornstalk cores air dried were mixed and covered completely with 15 g mixed salt in 50 ml porcelain crucibles, respectively. The mixtures were put into a muffle furnace and heated in air to 800 °C with a heating rate of 10 °C min⁻¹ and maintained at 800 °C for 3 h. Then, the muffle furnace was naturally cooled to room temperature. The products were washed with 1 M HCl solution and distilled water for several times and collected after drying at 60 °C for 12 h in an oven. The as-obtained samples were denoted as HPCS-3, HPCS-2, and HPCS-1 respectively. In addition, the comparison sample HPCS-3N was also prepared in a tubular furnace under nitrogen flow (100 mL min⁻¹) at the same conditions with HPCS-3. Moreover, cornstalk cores without immersing in the saturated salt solution were also directly mixed with solid salt and calcinated with the same heating profile as HPCS-3 and the sample is denoted as HPCS-3n. To investigate the influence of the reaction time, the samples HPCS-3-2h and HPCS-3-1h were also fabricated as the holding time were set to 2 and 1 h respectively. The carbon yields of the products at different reaction conditions were summarized in Table 1.

Table 1 Carbon yields under different producing conditions.

Samples	biomass (g)	Salt (g)	primary carbon product (g)	Carbon yields (%)
HPCS-1	1	15	0.225	22.50
HPCS-2	2	15	0.375	18.75
HPCS-3	3	15	0.450	15.00
HPCS-3n	3	15	0.480	16.00
HPCS-3N	3	15	0.710	23.60
HPCS-3-1h	3	15	0.693	23.10
HPCS-3-2h	3	15	0.680	22.67

2.2. Material characterizations

Simultaneous thermogravimetry and differential scanning calorimeter analyses were performed using a synchronous thermal analyzer (STA 449 F3) in the range of 40-1000 °C at a heating rate 10 °C min⁻¹ under air atmosphere. Field emission scanning electron microscopy (FESEM; Quanta FEG 250) and transmission electron microscopy (TEM; JEM-2100) were used to examine the morphology and micro-structure of the samples. Atomic force microscopy (AFM) imaging was scanned using a Bruker Dimension Icon with ScanAsyst. X-ray diffraction (XRD; Bruker D8 Advance) equipped with Cu K α radiation ($\lambda=1.5418$ Å) was employed to investigate the crystallographic structure of the samples. X-ray photoelectron spectroscopy (XPS) measurements were performed using Perkin-Elmer PHI-5700 ESCA System multifunctional photoelectron spectrometer with Al K α radiation (1486.6 eV) to study the surface chemical species of the samples. NOVA 2000e physisorption analyzer was employed to investigate the porous texture of the samples through the nitrogen adsorption/desorption isotherm measured at 77 K. The samples were degassed at 200 °C under vacuum for at least 6 h before measurements. The specific surface area of the carbon materials was calculated by the Brunauer-Emmett-Teller (BET) equation from the nitrogen adsorption data in the relative pressure (P/P_0) of 0.03-0.30. The pore size was calculated using the nonlocal density functional theory (NLDFT) equilibrium model for cylinder/slit pores from the N₂ desorption branch isotherms. The total pore volume (V_{total}) was determined at a relative pressure $P/P_0=0.990$ and the micro-pore volume (V_{micro}) using the t-plot method. A 4-point probes resistivity measurement system (RTS-9) was used to measure the electrical conductivity of the samples.

2.3. Electrochemical measurements

The electrochemical properties of the obtained HPCS were tested using 1 M H₂SO₄ solution as

electrolyte. For electrode preparation, a 80 wt % amount of HPCS, 15 wt % of acetylene black and 5 wt % of PTFE binder were mixed in ethanol. Then the paste was painted onto the current collectors of stainless steel mesh and dried at 60 °C for 6 h. The mass loading of active material on each electrode is 3 mg and the area painted is 1 cm². In a three-electrode system, a platinum foil and Ag/AgCl electrode was employed as the counter and reference electrode respectively. The cyclic voltammetry (CV), the galvanostatic charge/discharge (GCD) and the electrochemical impedance spectroscopy (EIS) tests were performed by a Zahner (Germany) electrochemical work station at room temperature. CV tests were investigated between 1 and 0 V (vs. Ag/AgCl) at different scan rates and GCD was tested at different current densities. EIS was operated in the frequency range from 100 kHz to 0.1 Hz with 5 mV voltage amplitude. The symmetrical supercapacitor was assembled using two same HPCS-3 electrodes and tested in a two-electrode system by the LAND series battery test system.

Specific capacitance for a single electrode in a three-electrode system and a supercapacitor in a two-electrode system was calculated from the discharge curves according to the following equations respectively:

$$C_g = \frac{I\Delta t}{m\Delta V} \quad (1)$$

$$C_g = \frac{2I\Delta t}{m\Delta V} \quad (2)$$

where I(A) is the discharge current, Δt (s) denotes the discharging time, m(g) is the mass of the electro active materials of one carbon electrode and ΔV is the potential change within the discharge time Δt .

The specific energy density (E_{cell}) and specific power density (P_{cell}) for symmetrical supercapacitors were also calculated based on the following formulas:

$$E_{cell} = \frac{C_g \Delta V}{8 \times 3.6} \quad (3)$$

$$P_{cell} = \frac{E_{cell}}{\Delta t} \quad (4)$$

where C_g is the specific capacitance of the symmetrical supercapacitor calculated from the discharge curves based on the total mass of electro active materials in two electrodes, ΔV is the voltage range and Δt is the discharge time.

3. Results and discussion

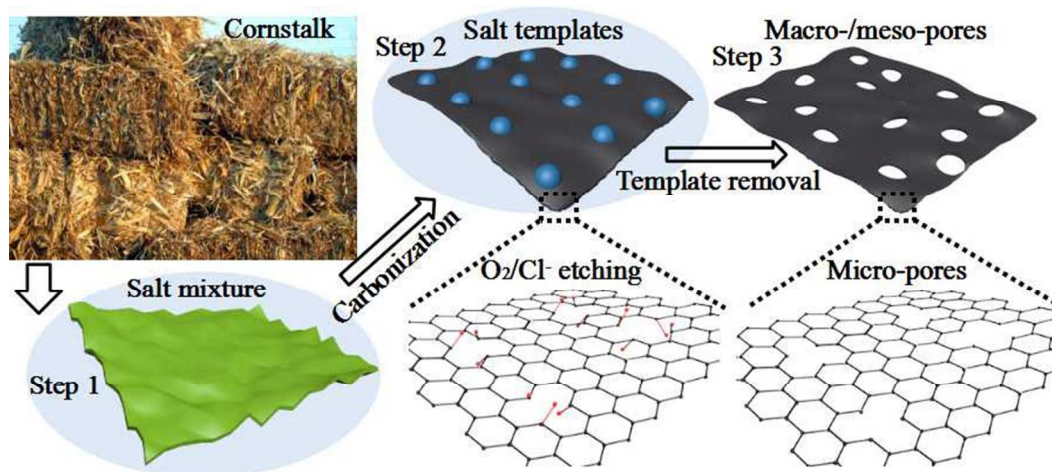


Fig. 1 Schematic illustration of the synthesis process for HPCS from cornstalk under air atmosphere.

Fig. 1 shows the schematic diagram for the preparation of HPCS in air. Firstly, the precursor are mixed and covered with salt. Afterwards, the mixture was calcinated in a muffle furnace at 800 °C for 3 h. After removing the salt, HPCS were obtained. In order to study the decomposition of cornstalk in air, the TG and DSC measurements were performed under air atmosphere. From TG and DSC curves in Fig. 2(a, b), the first weight loss of cornstalk (~58 %) ends at ~320 °C, which corresponds to the moisture elimination and the primary carbonization. The continuous weight loss (~35 %) in the range of 320-470 °C is due to the elimination of the volatile matters and the burning of the carbonized product. As the temperature is above 470 °C,

the weight loss is almost reach the maximum (~93 %), suggesting that the carbonized product has been burned completely without salt protection and the majority of the biomass was lost at 470°C due to the uncontrolled reaction with the O₂ in the air atmosphere. By comparison, for the mixture of cornstalk and salt (1:3, mass ratio), the weight loss (~6 %) in the range 40-320 °C and (~3 %) 320-470 °C results from the weight loss of the biomass because the molten salt experiences no obvious weight loss before 669 °C. Therefore, the ~9 % overall weight loss accounts for ~36 % weight loss of the cornstalk in the mixture, which is much less than the ~93 % weight loss without salt protection. When the temperature was further increased to above 669 °C (the melting point of the salt mixture), the weight loss is mainly due to the evaporation of molten salt according to the TG curve.

The TG result indicates the salt protection could prevent the biomass from burning and a possible mechanism was proposed as follow. Before the molten point (669 °C) (Step 1, in Fig.1), the salt firstly acts the solid sealing agent on the carbonized product (Fig. S1(a)) to prevent the reaction with O₂ in the air. At the meantime, the biomass would experience a carbonization process through the steady transition from sp³C-X (X: e.g., C, O, H) bonds to the aromatic sp² C-C bonds, which forms the skeleton of the carbon structure⁴⁸. When the temperature is elevated to 669 °C, the liquid phase salt would diffuse into the carbonized product and serve as liquid sealing agent to prevent the carbon product from burning. As the reaction temperature was held at 800 °C (Step 2, in Fig.1), the molten salt media can provides high energy Cl⁻ ions to etch the carbon structures and form micro- and meso- pore structures^{8, 48}. In addition, the molten salt diffused into the tissue structure can function as “cutting” regent to prohibit the formation of van der Waals force, which suppresses the stacking of the sp² coordinated carbon layers along the C-axis and finally forms the layer structure of the final product⁴⁸. When the temperature is cooled

down below the melting point (Step 2, in Fig.1), the salt clusters and nanosized salt particles would precipitate from molten state and embed in the carbon texture due to the strong binding interactions, which could serve as template to form the lots of meso- and macro-pores with relative greater diameters (Fig. S1(b))⁴⁷. In addition, with the elongating the calcination time, oxygen in air would penetrate into molten salt to etch the carbon sheets (Step 2, in Fig.1) due to the fact that carbon itself is highly reductive at high temperatures from a thermodynamic viewpoint⁴⁸, generating amounts of micro-pores on carbon sheets. Thus, as the salt media was removed, two-dimensional carbon sheets with hierarchical porous structure were finally formed (Step 3, in Fig.1).

Fig. 2(c) shows the FESEM image of HPCS-3, revealing that a large amount of thin carbon sheets were obtained. The high resolution FESEM image in Fig. 2(d) shows that the thickness of HPCS-3 carbon sheets is less than 10 nm. The AFM image of HPCS-3 and the curve based on the thickness of the sample vs. length were further shown in Fig. 2(e, f), revealing that the thickness of an individual HPCS-3 carbon sheet is ~4.6 nm. Detailed transmission electron microscopy (TEM) examinations were performed and lots of macro-pores with diameter of 50~200 nm and meso-pores with diameters of 2-7 nm obviously could be observed in Fig. 2(g, h). In addition, the inset of selected area electron diffraction (SAED) image in Fig. 2(i) displays diffused rings, suggesting that HPCS-3 have a disordered structure with amounts of defects and discontinuous graphite stripes as shown in Fig. 2(i).

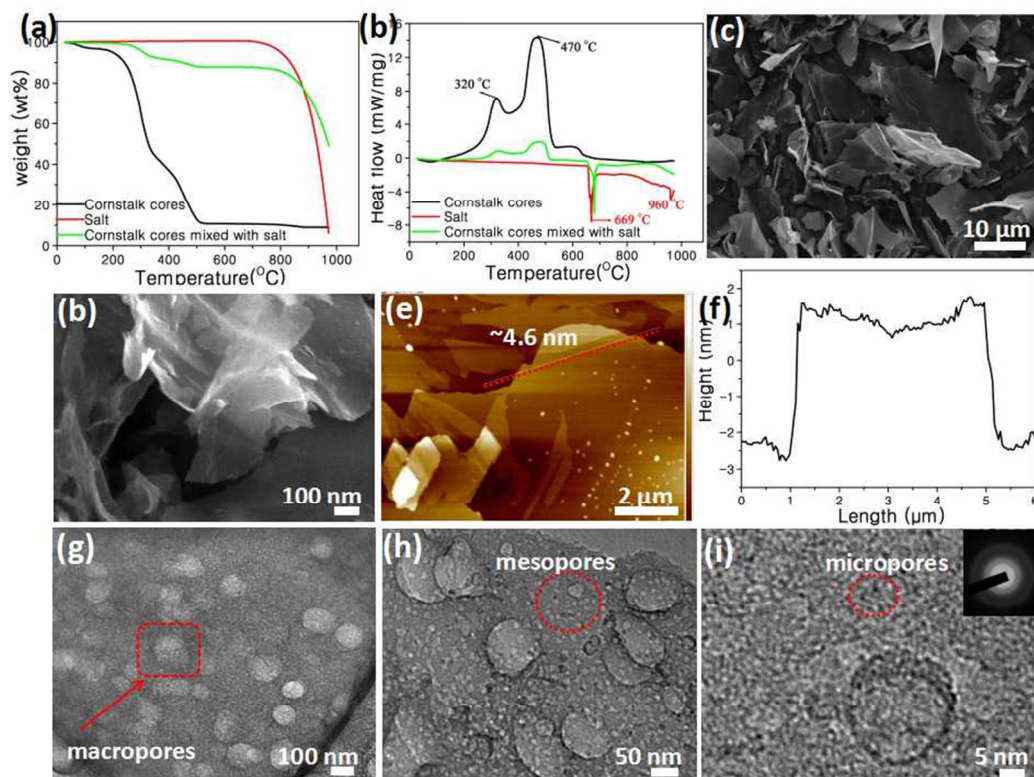


Fig. 2 (a, b) TG and DSC diagrams of cornstalk, salt and the cornstalk/salt mixture, (c, d) FESEM images, (e) AFM image, (f) height of the sample vs. length and (g-i) TEM images of HPCS-3 (the inset is the SAED image).

Additionally, the low resolution FESEM images of the comparison samples of HPCS-2, HPCS-1, HPCS-3n and HPCS-3N were shown in Fig. 3 (a, c, e), in which lots of carbon sheets were also observed. The high resolution images in Fig. 3(b, d, f) show that the thickness of the comparison samples are around 25 nm (HPCS-2), 30 nm (HPCS-1), 50 nm (HPCS-3n) and 50 nm (HPCS-3N). These results suggest that besides molten salt, oxygen penetrated into the molten salt would also etch the formed carbon sheets, which is favorable for the formation of the ultra-thin carbon layer structure^{5,48}.

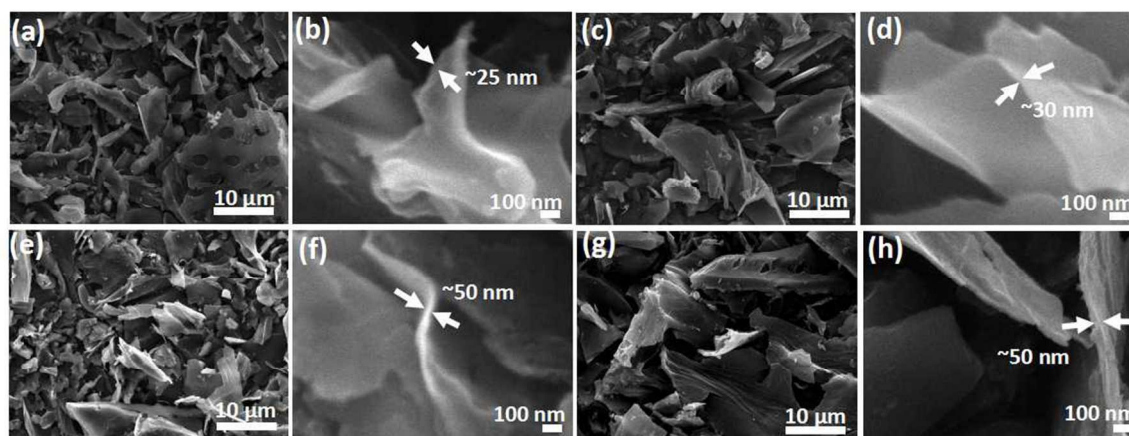


Fig. 3 FESEM images of HPCS-2 (a, b), HPCS-1 (c, d), HPCS-3n (e, f) and HPCS-3N (g, h).

X-ray diffraction (XRD) patterns in Fig. 4(a) show that the samples HPCS-3, HPCS-2, HPCS-1, HPCS-3n and HPCS-3N exhibit two broad peaks around 23.6° corresponding to the (002) and 44° corresponding to the (100) plane of graphite, suggesting the formation of carbon product with a limited graphitization degree, consistent with the TEM observations. To further analyze the composition of the samples, the corresponding XPS survey spectra of all samples are shown in Fig. 4(b), displaying two peaks at binding energy of 284.6 and 532.4 eV assigned to C1s and O1s. The content of C1s and O1s are calculated to be HPCS-3 (89.32 and 10.68 at%), HPCS-2 (89.82 and 10.18 at%), HPCS-1 (90.88 and 9.12 at%), HPCS-3n (87.15 and 12.85 at%) and HPCS-3N (88.84 and 11.16 at%). The high-resolution C1s spectra of HPCS-3 in Fig. 4(c) further shows that three peaks located at 284.5, 285.8 and 287.8 eV are attributed to C=C, C-O and C=O groups, respectively. The corresponding high-resolution O1s spectra of HPCS-3 in Fig. 4(d) further evidence the presence of C-O and C=O groups at 532.8 and 531.6 eV. These rich oxygenated functional groups can contribute pseudo capacitance during the charge/discharge process based on the reversible oxidation/reduction^{9, 10, 21, 43}.

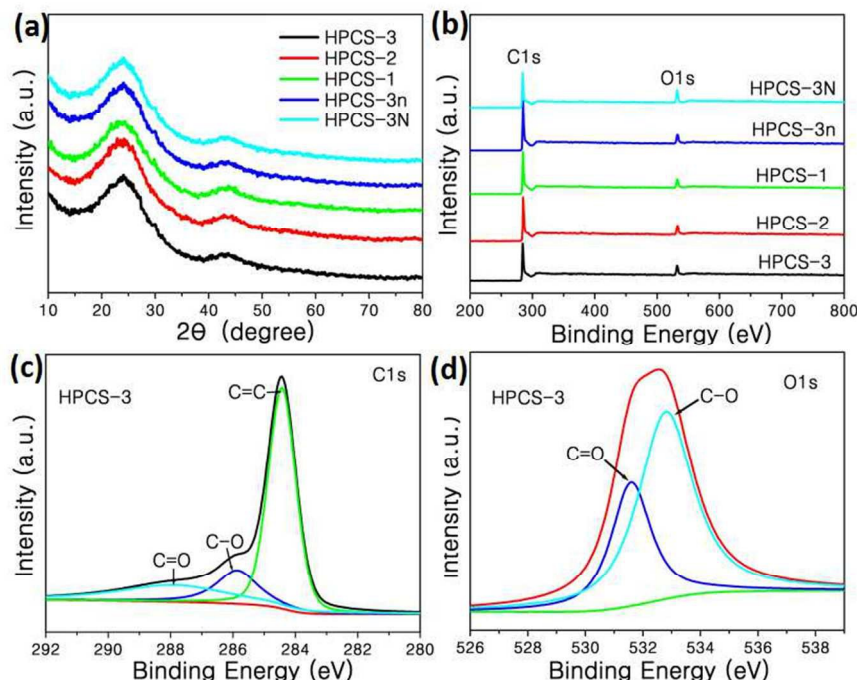


Fig. 4 (a) XRD patterns and (b) XPS survey spectra of HPCS-3, HPCS-2, HPCS-1, HPCS-3n and HPCS-3N, (c) C1s and (d) O1s spectra of HPCS-3.

The pore structures of the carbon samples were examined by nitrogen adsorption/desorption measurements at 77 K. The nitrogen adsorption/desorption isotherms of all samples in Fig. 5(a) show the apparent hysteresis loops in the medium relative pressure, demonstrating the presence of abundant meso-pores. Moreover, their nitrogen adsorption/desorption isotherms also display a relatively high nitrogen uptake in the low relative pressure, suggesting that along with meso-pores large amounts of micro-pores were generated as well. In addition, a slight upward tendency at high relative pressure ($P/P_0=0.95-1.0$) indicates the existence of macro-pores. By comparison, HPCS-3 obviously exhibits the highest nitrogen uptake among all the samples. Fig. 5(b) shows the pore size distribution calculated by DFT model, revealing that the samples have the maximal peaks at 1-2 and 2-7 nm belonging to micro-pores and meso-pores. The SSA and pore structure parameters of the as-obtained samples are surmised in Table 2. HPCS-3 exhibits the highest SSA

of $1588 \text{ m}^2 \text{ g}^{-1}$, total pore volume of $1.018 \text{ cm}^3 \text{ g}^{-1}$ and the meso-pore volume of $0.573 \text{ cm}^3 \text{ g}^{-1}$, confirming its optimized hierarchical pore structure.

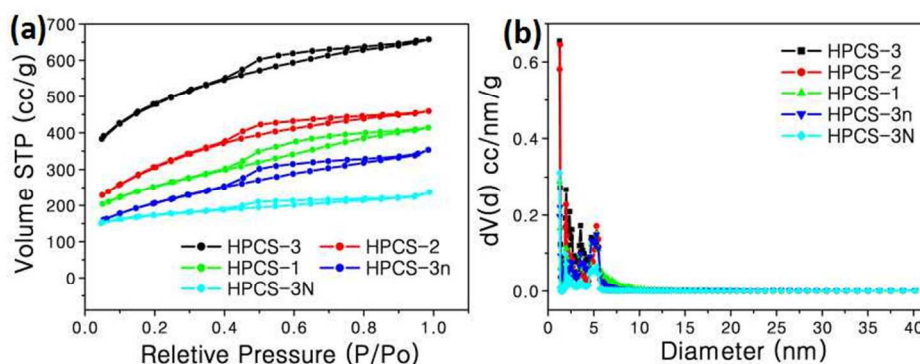


Fig. 5 (a) N_2 adsorption/desorption isotherms and (b) DFT pore size distribution of HPCS-3, HPCS-2, HPCS-1, HPCS-3n and HPCS-3N.

Table 2 Pore textural properties of the as-obtained porous carbon materials

Samples	$S_{\text{total}} (\text{m}^2 \text{ g}^{-1})$	$V_{\text{total}} (\text{cm}^3 \text{ g}^{-1})$	$V_{\text{micro}} (\text{cm}^3 \text{ g}^{-1})$	$V_{\text{meso}} (\text{cm}^3 \text{ g}^{-1})$
HPCS-3	1588	1.018	0.445	0.573
HPCS-2	1066	0.711	0.188	0.523
HPCS-1	864	0.640	0.174	0.466
HPCS-3n	847	0.589	0.386	0.203
HPCS-3N	655	0.431	0.278	0.153
HPCS-3-1h	551	0.366	0.199	0.167
HPCS-3-2h	712	0.547	0.212	0.335

S_{total} : total BET specific surface area; V_{total} : total pore volume; V_{micro} : micro-pore volume; V_{meso} : meso-pore volume.

These pore structure results demonstrate that the etching effect of oxygen and the salt template-directing reagent play crucial roles on the formation of the hierarchical pores in carbon sheets (Fig. 1). The SSA of HPCS-3N prepared in the N_2 atmosphere is $655 \text{ m}^2 \text{ g}^{-1}$, which is obviously lower than that of HPCS-3 ($1588 \text{ m}^2 \text{ g}^{-1}$), confirming that oxygen in air is indispensable for generating pores in the carbon sheets. On the other hand, salt itself can act as the template-directing agent embedded in the carbon structures to generate hierarchical pores⁴⁷.

This template-directing role depends on the homogeneous mixing of the salt within the substructure of cornstalk. The SSA of sample HPCS-3n derived from the cornstalk without immersion in the saturated salt solution is $847 \text{ m}^2 \text{ g}^{-1}$, lower than that of HPCS-3, demonstrating that the homogeneous infiltration of the salt within the substructure of cornstalk is also crucial for the formation of hierarchical pores. Additionally, the mass ratio of the cornstalk/salt could also regulate SSA and porosity of the samples. With decreasing salt dosages, the SSA of HPCS-1, HPCS-2 and HPCS-3 is increased from 864 to 1066 and $1588 \text{ m}^2 \text{ g}^{-1}$ with total pore volume from 0.640, 0.711 to $1.018 \text{ cm}^3 \text{ g}^{-1}$. For the low salt dosage (HPCS-3), the insufficient sealing effect leads to more air penetrating into the reaction system, leading to the highest pore volume and SSA.

The effect of calcination time on the pore structures of the carbon samples was also investigated. The FESEM images of HPCS-3-2h and HPCS-3-1h were shown in Fig. S2. The corresponding nitrogen sorption isotherms in Fig. S3 show that HPCS-3-2h and HPCS-3-1h obviously exhibit lower nitrogen uptake than HPCS-3. The values of the SSA and total pore volume in Table 2 reveal that the SSA of HPCS-3-2h and HPCS-3-1h was decreased to 712 and $551 \text{ m}^2 \text{ g}^{-1}$ from $1588 \text{ m}^2 \text{ g}^{-1}$ of HPCS-3 and the total pore volume from $1.018 \text{ cm}^3 \text{ g}^{-1}$ to 0.547 and $0.366 \text{ cm}^3 \text{ g}^{-1}$ respectively. These results confirmed that with the longer calcination time, the better infiltration of molten salt could be achieved and the homogeneous salt distribution within the carbon skeleton leads to better etching effect and thinner layered carbon structures. In addition, more air will penetrate the molten salt to etch the carbon sheets, which could also contribute to the micro-pore structure and high SSA with longer reaction time.

In addition, the electronic conductivity of the carbon samples was further measured by a 4-probe method. The results show that the samples exhibit relatively similar conductivity of 4.38 S

cm^{-1} (HPCS-3), 4.18 S cm^{-1} (HPCS-2), 4.13 S cm^{-1} (HPCS-1), 3.61 S cm^{-1} (HPCS-3n), 3.60 S cm^{-1} (HPCS-3N), 3.75 S cm^{-1} (HPCS-3-2h) and 2.98 S cm^{-1} (HPCS-3-1h). These results suggest that the conductivity of the carbon samples were not greatly affected by the contents of oxygen and the pore structures in the carbon sheets, indicating the effective protection of the carbon product due to the molten salt media.

HPCS with above excellent characteristics should be ideal electrodes for supercapacitors. A three-electrode system was firstly used to test the electrochemical performances of the samples in $1 \text{ M H}_2\text{SO}_4$ aqueous solution. As shown in Fig. 6(a) and Fig. S4(a), the cyclic voltammetry (CV) curves of all samples at a scan rate of 80 mV s^{-1} take roughly rectangular shapes with obvious humps at around $0.2\sim 0.6 \text{ V}$ attributed to the redox reactions resulted from oxygenated functional groups in HPCS^{51, 55}, suggesting their combined double layer capacitance and pseudo-capacitance. By comparison, the CV profile of HPCS-3 obviously shows the largest encircled area, indicating its best capacitance characteristic. Fig. 6(b) and Fig. S4(b) display the corresponding galvanostatic charge-discharge (GCD) curves of all samples at 1 A g^{-1} , suggesting that all samples take imperfectly symmetrical triangular shapes which are similar to those of biomass-derived carbons in the literatures^{49, 51-55} and HPCS-3 apparently shows the longest charge-discharge time. Calculated from the GCD curves, the specific capacitance of HPCS-3 is 407 F g^{-1} at 1 A g^{-1} , obviously higher than 199 F g^{-1} (HPCS-2), 171 F g^{-1} (HPCS-1), 153 F g^{-1} (HPCS-3n), 125 F g^{-1} (HPCS-3N), 135 F g^{-1} (HPCS-3-2h) and 126 F g^{-1} (HPCS-3-1h) (Fig. S4). As for HPCS-3, the ultra-thin sheet structure facilitates the ion transportation⁴⁶, the highest SSA of $1588 \text{ m}^2 \text{ g}^{-1}$ provides sufficient electrode-electrolyte interfaces for charge capacity⁴⁶ and the largest amounts of meso-pores benefit for the retention and immersion of the electrolyte, providing convenient channels for electrolyte penetration and ion transportation^{45, 46}. Meanwhile,

the rich oxygen-containing functional groups introduce pseudo capacitance into the sample^{9, 10, 21, 43}. Therefore, compared with many reported carbon electrodes illustrated in Table 3, HPCS-3 (407 F g⁻¹) highlights superb gravimetric specific capacitance.

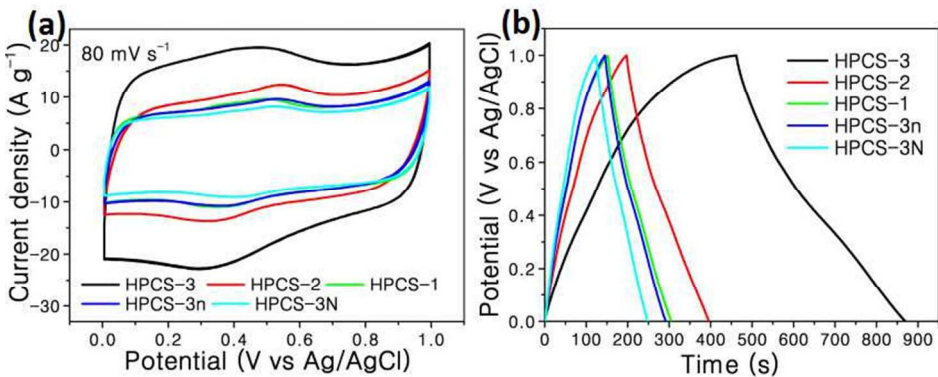


Fig. 6 Electrochemical performance in a three-electrode system in 1.0 M H₂SO₄ aqueous solution (a) CV curves at 80 mV s⁻¹ and (b) GCD curves at 1 A g⁻¹ of HPCS-3, HPCS-2 , HPCS-1, HPCS-3n and HPCS-3N.

Table 3 The comparison of electrochemical performance between HPCS-3 and the carbon electrodes reported in the literatures.

Carbon precursors	Gravimetric specific capacitance (F g ⁻¹)	Electrolyte	Current density	Reference
Hydro-char	279	1 M H ₂ SO ₄	0.1 A g ⁻¹	54
Bagasse wastes	320	6 M KOH	0.5 A g ⁻¹	55
Corncob	401	1 M H ₂ SO ₄	0.5 A g ⁻¹	56
Defatted soybean	260	1 M H ₂ SO ₄	0.2 A g ⁻¹	57
Chitosan	252	6 M KOH	0.5 A g ⁻¹	49
Tobacco rods	286	6 M KOH	0.5 A g ⁻¹	53
Peanut shell	160	6 M KOH	1 A g ⁻¹	51
Wheat flour	383	1 M H ₂ SO ₄	0.25 A g ⁻¹	58
Willow catkins	298	6 M KOH	0.5 A g ⁻¹	59
Cornstalk cores	407	1 M H ₂ SO ₄	1 A g ⁻¹	This work

Due to its hierarchical pore structure providing electrolyte reservoirs and convenient ion transportation channels which ensures effective accessibility of electrolyte ions to the electrode

surface at high charge/discharge rates, HPCS-3 shows rectangular-shaped CV curves at different scan rates from 20 to 200 mV s^{-1} (Fig. 7(a)) and the corresponding GCD curves at different current densities from 1 to 20 A g^{-1} (Fig. 7(b)), demonstrating an ideal capacitive behavior and remarkable rate performance. Compared with the CV and GCD curves of the comparison samples HPCS-2 and HPCS-1 in Fig. 7 c-f, HPCS-3n and HPCS-3N in Fig. 8 and HPCS-3-2h and HPCS-3-1h in Fig. S4, the excellent electrochemical performances of HPCS-3 are further demonstrated.

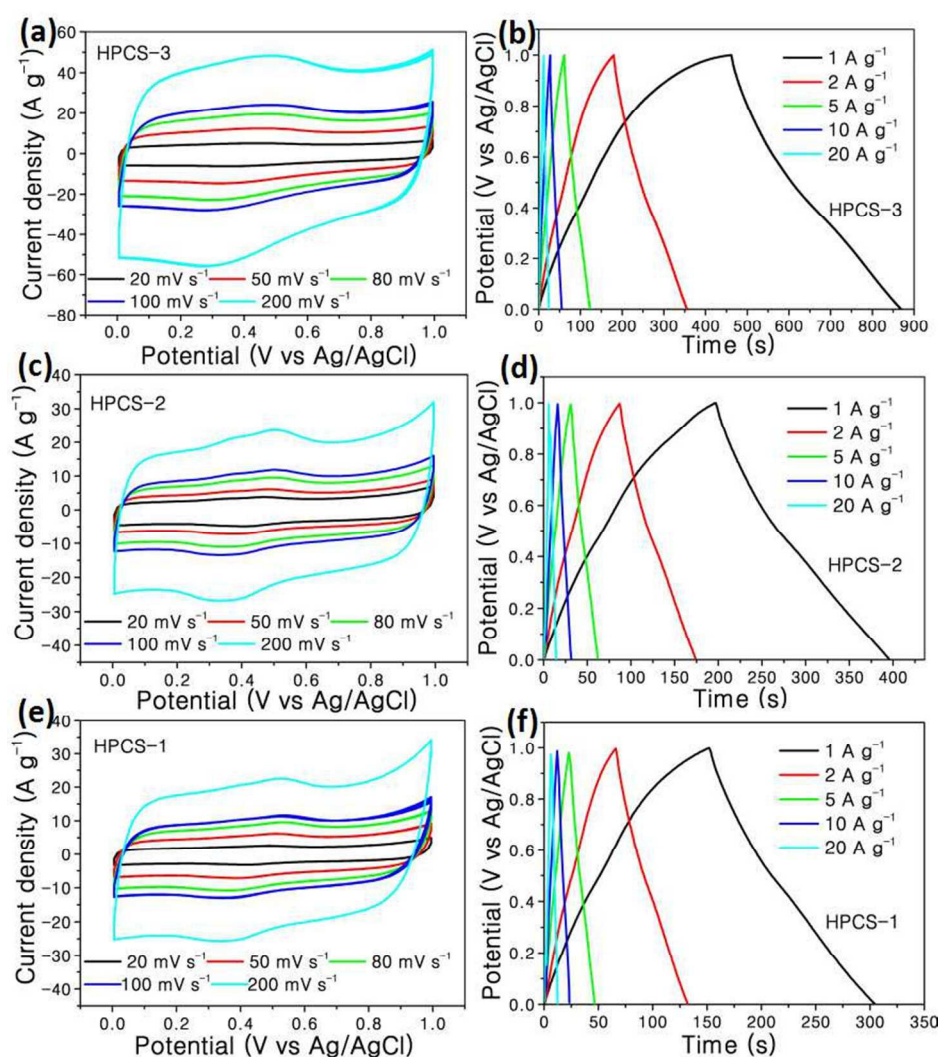


Fig. 7 CV curves at different scan rates and GCD curves at different current densities of HPCS-3, HPCS-2, and HPCS-1 in a three-electrode system in 1.0 M H_2SO_4 aqueous solution.

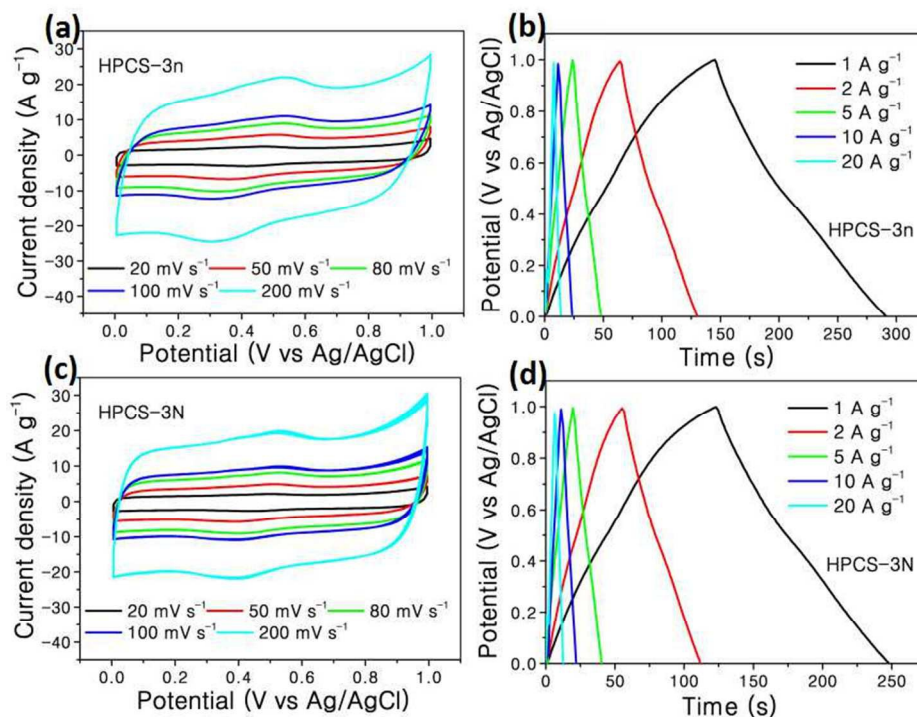


Fig. 8 CV curves at different scan rates and GCD curves at different current densities of HPCS-3n and HPCS-3N in a three-electrode system in 1.0 M H_2SO_4 aqueous solution.

The specific capacitance versus current density of all samples shown in Fig. 9(a) and Fig. S5(a) shows that HPCS-3 still offers a high specific capacitance of 246 F g^{-1} even at 20 A g^{-1} (60.4 % capacitance retention), suggesting good rate capacity. The superb rate capacity benefits from the low ion transport resistance and short diffusion distance during the charge-discharge process^{46, 55}. To further investigate the electrolyte ions diffusion and transport kinetics, electrochemical impedance spectroscopy (EIS) was further performed. Fig. 9(b) and Fig. S5(b) display the Nyquist plots of all carbon electrodes. The corresponding fitting equivalent circuit adopted to fit the plots was inserted in Fig. 9(b). The steepest linear curve of HPCS-3 in the low frequency region means an ideal capacitive performance and a low equivalent series resistance of ion diffusion and migration^{13, 46, 55}. The smallest semicircle diameter of HPCS-3 in the high frequency region corresponds to a low charge transfer resistance⁵⁵ of R_{ct} $1.06 \text{ } \Omega$ which is

apparently lower than 3.02 Ω (HPCS-2), 3.86 Ω (HPCS-1), 4.97 Ω (HPCS-3n), 7.39 Ω (HPCS-3N), 3.95 Ω (HPCS-3-2h) and 6.37 Ω (HPCS-3-1h). The thinner two dimensional sheet structure of HPCS-3 permit the accessibility of ions on the active site from both sides which can shorten the length of ion transportation⁴⁶. Moreover, the abundant macro- and meso-pores in carbon sheets supply facile transport channels for the electrolyte transportation^{13, 46, 55}, which can reduce the charge transfer resistance, thus leading to the lowest R_{ct} . Additionally, HPCS-3 also exhibits the shortest Warburg resistance section of Z_w 0.36 Ω lower than 0.38 Ω (HPCS-2), 0.41 Ω (HPCS-1), 0.52 Ω (HPCS-3n), 0.53 Ω (HPCS-3N), 0.51 Ω (HPCS-3-2h) and 0.57 Ω (HPCS-3-1h), which means a short ion diffusion path permitting more efficient access of the electrolyte ions to the electrode surface¹⁸.

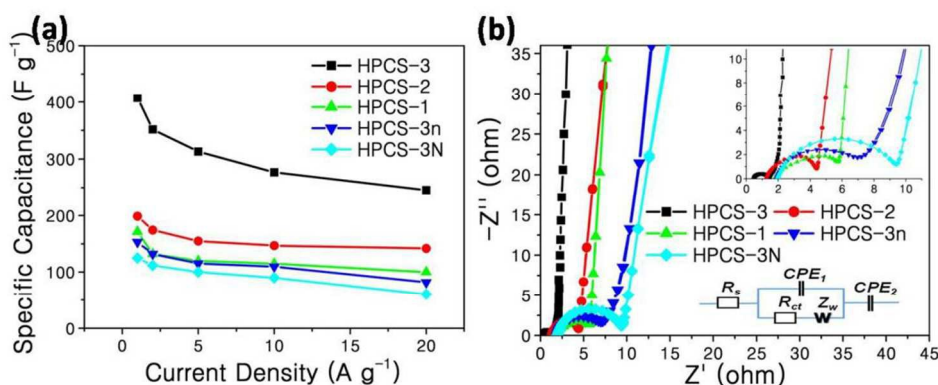


Fig. 9 (a) Specific capacitance at different current densities and (b) Nyquist plots of HPCS-3, HPCS-2, HPCS-1, HPCS-3n and HPCS-3N in a three-electrode system in 1.0 M H_2SO_4 aqueous solution (The inset was the fitting equivalent circuit).

Symmetric supercapacitor was further assembled by employing two HPCS-3 electrodes and tested in a two-electrode system in 1 M H_2SO_4 aqueous electrolyte. Fig. 10(a) shows the GCD curves of the symmetric device at different current densities from 0.5 to 20 $A g^{-1}$. The symmetrical triangular charge-discharge shapes with negligible IR drop over a wide range of

current density suggest that the device exhibits a typical electrochemical capacitor behavior. Calculated from the GCD curves, the specific capacitance of the device was 413 F g^{-1} at 0.5 A g^{-1} and 313 F g^{-1} at 1 A g^{-1} , demonstrating an outstanding capacitance performance. The high specific capacitance of the supercapacitor is ascribed to the excellent characteristics of HPCS-3 electrode including large SSA which guaranteed adequate interfaces for ion and charge capacity, the multilevel hierarchical pores which guaranteed the efficient utilizations of the electrode surface and the rich O-doping which contributed pseudo capacitance based on the reversible oxidation/reduction^{9, 21, 43}. Therefore, the capacitance value of HPCS-3//HPCS-3 supercapacitor is even superior to that of previously reported symmetric supercapacitors employing carbon materials derived from other biomass in aqueous electrolyte, such as coconut-shell (268 F g^{-1} at 1 A g^{-1})⁶⁰, chicken eggshell membranes (284 F g^{-1} at 0.2 A g^{-1})⁶¹, protein (390 F g^{-1} at 0.25 A g^{-1})²⁵, and microorganism (305 F g^{-1} at 0.2 A g^{-1})⁶². The specific capacitance at different current densities is further summarized in Fig. 10(b), revealing relatively small degradation in the specific capacitance when current density increases from 2 A g^{-1} (286 F g^{-1}) to 10 A g^{-1} (272 F g^{-1}), demonstrating remarkable rate performance. Generally, the excellent rate capacity of the device is close associated with the low polarization effects of the electrode, which is governed by the high electron conductivity of the electrodes and rapid electrolyte diffusion⁴⁶. For HPCS-3, the higher electronic conductivity (4.38 S cm^{-1}) guarantees the low internal resistance of the electrodes. Meanwhile, the thin two dimensional carbon sheets with abundant macro- and mesopores supply short transfer pathway and facile transport channels for ions, which is favorable for the mass transportation and reducing the charge transfer resistance, leading to the minimized polarization effects and good rate capacity of the device.

Additionally, the energy and power densities of the symmetric supercapacitor were also

calculated and the Ragone plot shown in Fig.10(c) depicts that the energy density for the device is 14.35 Wh kg^{-1} at a power density of 123.9 W kg^{-1} , superior to the commercial active carbon ($5\text{--}8 \text{ Wh kg}^{-1}$). Cycle life test further conducted (Fig.10(d)) reveals that the device HPCS-3//HPCS-3 also shows a long-term cycling stability with 92.6 % of the initial capacitance (278 F g^{-1}) retention after 20000 cycles at a current density of 5 A g^{-1} , which is crucial for the devices in the energy storage applications.

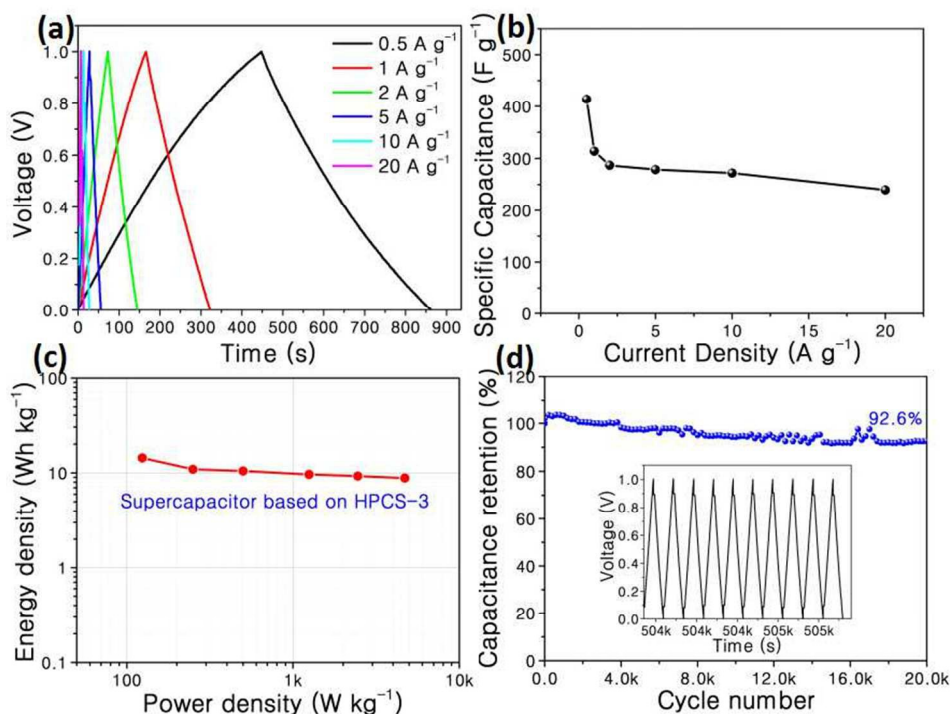


Fig. 10 Electrochemical performances of the symmetrical supercapacitor in a two-electrode system in 1.0 M H₂SO₄ aqueous solution. (a) GCD curves and (b) specific capacitance of HPCS-3//HPCS-3 at different current densities, (c) Ragone plot, (d) cycling stability at 5 A g^{-1} up to 20000 cycles (the inset is the GCD curves of the last ten cycles).

4. Conclusions

In summary, HPCS were directly prepared from biomass in air at 800°C . Compared with the conventional chemical activation method in the protection of inert gas, this technique is more

convenient for operation and avoid the consumption of inert protection gas as well as the use of complicated equipments. In addition, it adopts non-toxic chloride salt as reaction media, which simultaneously integrates the carbonization and activation processes in producing the final products. The as-obtained HPCS-3 exhibits excellent characteristics consisting of ultra-thin carbon sheets facilitating the fast ion transfer, abundant macro- and meso-pores furnishing unblocked ion diffusion channels, high SSA providing sufficient electrode/electrolyte interfaces for charge capacity and rich O doping introducing pseudo capacitance. As electrode materials for supercapacitors, HPCS-3 exhibits a high specific capacitance of 407 F g^{-1} at 1 A g^{-1} in a three-electrode system. Moreover, the assembled symmetrical HPCS-3//HPCS-3 device also exhibits an outstanding specific capacitance of 413 F g^{-1} at 0.5 A g^{-1} , good rate capacity and a long-term cycling stability with only 7.4% loss after 20 000 cycles at 5 A g^{-1} in a two-electrode system. These results further suggest that this green and scalable route represents a promising strategy to prepare porous carbon materials in the field of energy storage, catalysis or gas storage systems.

Acknowledgments

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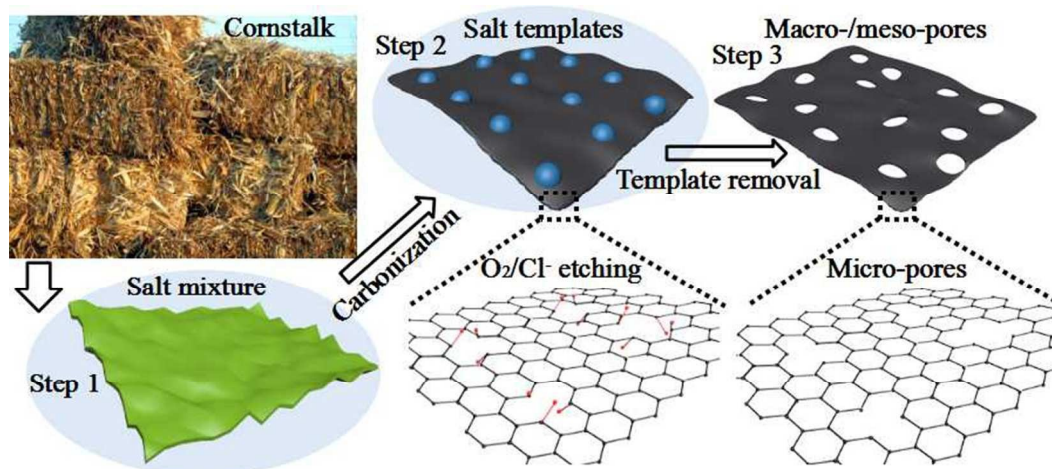
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Table of Content



A green route is developed to prepare hierarchical porous carbon sheets (HPCS) from biomass directly under air atmosphere instead of inert gas protection. As supercapacitor electrodes, the HPCS with large SSA of $1588 \text{ m}^2 \text{ g}^{-1}$ demonstrate an excellent specific capacitance of 407 F g^{-1} at 1 A g^{-1} and the symmetrical supercapacitor in two-electrode system also exhibits an outstanding specific capacitance of 413 F g^{-1} at 0.5 A g^{-1} .